Fiscal Year:	FY 2023	**Task Last Updated:**	FY 05/25/2023
PI Name:	Masson, Patrick H. Ph.D.		
Project Title:	Using Brachypodium distachyon to Study the Molecular Mechanisms That Underlie Directional Root-Growth Responses to Low-Speed 2-Dimensional Clinorotation		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Zip Code:	53706-1580	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	08/15/2018	**End Date:**	12/31/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2022 per NSSC information (Ed., 6/1/23). NOTE: End date changed to 08/14/2022 per NSSC information (Ed., 5/17/22).		
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COI Name (Institution):			
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Task Description:	As key components of bioregenerative life-support systems, plants will be exposed to multiple stresses associated with spaceflight. Experiments carried out in Space have dramatically improved our understanding of plant responses to microgravity. In turn, this novel knowledge has allowed the design of better culturing methods that enable plant development and reproduction in Space. Yet, to design even better growth strategies and develop optimized genotypes favorable to plant growth and productivity during spaceflight, we need to better understand the molecular machinery that contribute to the perception of, and response to gravity and/or microgravity. Unfortunately, the large cost associated with Space experimentation and limited accessibility to the resources needed for such experimentation, limit our ability to acquire this new knowledge. Consequently, utilization of devices that mimic microgravity is viewed as a viable alternative. The 2D-clinostat has often been utilized for this purpose. By constantly changing the orientation of biological specimens relative to gravity, these devices prevent gravity-vector detection, thereby triggering behavioral responses that are similar to those expressed under true microgravity. The 2D-clinostat has often been used to estimate the sensitivity of plant organs to different doses of gravi-stimulation. In this case, low doses of gravi-stimulation are provided at the beginning of the experiment. Stimulated seedlings are subsequently clinorotated for 4-5 hours, allowing for a curvature to develop in response to the initial stimulus. Yet, the data derived from these analyses are likely to be significant only if the clinorotation does not introduce artifacts of its own. In experiments involving Arabidopsis thaliana seedlings, we previously observed a strong clinostat effect on root skewing on hard agar surfaces. Follow-up experiments involving Brachypodium distachyon confirmed a clear impact of low-speed clinorotation on directional root-tip bending. These experiments also suggested an important role for the first cycle of clinorotation in setting up a specific direction of gravistimulation as long as it is provided for a sufficiently long time period. The general objective of this research project will be to investigate the effect of low speed clinorotation on Brachypodium root growth. Specific objectives include: Aim 1: Develop high throughput time-lapse imaging to describe the development of B. distachyon root curvature responses to clinorotation at high spatio-temporal resolution. Aim 2: Use transgenic B. distachyon seedlings that express auxin-level, auxin-function, and auxin-efflux carrier reporters to investigate the effect of clinorotation on auxin distribution and signaling along the root tip, and on the distribution of auxin transporters between flanks of clinorotated root tips. Aim3: Use a combination of Genome Wide Association Study (GWAS) and RNA Seq strategies to identify candidate genes that may contribute to root-tip response to clinorotation, and use reverse genetics to investigate the role(s) of such candidates and of gravitropism-related genes in the process. Overall, this project will improve our understanding of how low-speed clinorotation affects root growth behavior. While our observations suggest a need for greater caution in using a low-speed 2D clinostat as a microgravity-mimic system, they also suggest a novel method to investigate the molecular mechanisms that contribute to gravity sensing in plants, one of the main objectives of the Space Biology program. This project will introduce undergraduate students to the idiosyncrasies of scientific research in Space Biology, while also putting a strong emphasis on undergraduate and K-12 education and public outreach.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project improves our understanding of how low-speed clinorotation affects root growth behavior, while re-emphasizing a need for greater caution in using a low-speed 2D clinostat as a microgravity-mimic system. Our research resulted in the development of a novel method to investigate the molecular mechanisms that contribute to gravity sensing in plants, one of the main objectives of the Space Biology program. In the longer term, the basic information acquired during these studies should enable the design of plants that are better suited to function as key components of bioregenerative life support systems for long-term space exploration missions, either during travel under microgravity, or exposed to distinct gravity environments on exotic planets. They may also allow the development of novel crop cultivars with increased productivity in natural and/or agricultural ecosystems on Earth. With this project, we were able to introduce undergraduate students to the idiosyncrasies of scientific research in Space Biology, in addition to developing effective educational materials to introduce K-12 and undergraduate students, as well as the general public, to the difficult concepts of gravity perception by plants and its importance for agricultural production on Earth and during spaceflight.
Task Progress:	Plants have been recognized as essential companions for long-term space exploration missions, providing key functions in bioregenerative life-support systems, producing food, feed, fiber, biofuels, and biopharmaceuticals, and also ensuring the well-being of the astronauts traveling within the confines of a spacecraft. To make this happen, we should better understand how plants respond to the microgravity and fractional gravity environments encountered during spaceflight. While experiments carried out in space are optimal to investigate these responses, they suffer from prohibitive cost and limited access. An alternative solution used by investigators on the ground to mimic microgravity relies on devices such as the clinostats, which are instruments that continuously rotate the biological specimen along an axis, thereby averaging their orientation relative to gravity over time. Clinostats have been used extensively to estimate plant gravisensitivity, for instance. Yet, our preliminary experiments indicated that low-speed clinorotation may itself trigger a directional root-tip curvature. In this project, we have investigated the root-curvature response to low-speed clinorotation using Arabidopsis thaliana and Brachypodium distachyon seedlings as models. We showed that slow clinorotation triggers a root tip curvature whose direction is dictated by gravitropism during the first half-turn of clinorotation. We also showed that the angle of root tip curvature is modulated by the speed of clinorotation. Arabidopsis mutations affecting gravity susception (pgm) or gravity signal transduction (arg1, toc132) also affect the root-tip curvature response to low-speed clinorotation. Furthermore, low-speed vertical clinorotation also triggers a relocalization of the PIN3 auxin efflux facilitator to the lateral membrane of Arabidopsis root cap statocytes, a process that is accompanied by the creation of a lateral gradient of auxin across the root tip. Together, these observations support a role for gravitropism in modulating root-curvature responses to clinorotation. Finally, we showed that different Brachypodium distachyon accessions display distinct abilities to develop root-tip curvatures in response to low-speed vertical clinorotation, suggesting the possibility of using genome-wide association studies to further investigate this process. This investigation suggests a need for greater caution in the use of clinorotation to investigate gravisensitivity in plants and also provides a new tool to investigate the molecular mechanisms that modulate gravity sensing and signal transduction.

Bibliography Type:	Description: (Last Updated: 07/03/2024)
Articles in Peer-reviewed Journals	Su SH, Moen A, Groskopf RM, Baldwin KL, Vesperman B, Masson PH. "Low-speed clinorotation of Brachypodium distachyon and Arabidopsis thaliana seedlings triggers root tip curvatures that are reminiscent of gravitropism." Int J Mol Sci. 2023 Jan 12;24(2):1540. https://doi.org/10.3390/ijms24021540 PMID: 36675054; PMCID: PMC9861679 , Jan-2023